

Panel MS-AR(1): common intercept and linear trend

Full sample, seasonally adjusted real GDP per worker

1 Model

All countries and all years in the 2026-09-17 panel. No country random effects and no catch-up term.

$$y_{it} = a + g t + z_{it}, \tag{1}$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \tag{2}$$

$$s_{i,t+1} \sim \Pi(\cdot \mid s_{it}). \tag{3}$$

Three regimes. The unconditional mean of the cycle is restricted to $E[z] = 0$ (the median regime mean is not pinned at 0). σ and ρ switch, with $|\rho| < 0.99$. Standard errors are delta-method from a numerical Hessian.

2 Common intercept and linear trend, full sample

2026-09-17 SA panel, full sample, $y_{it} = a + gt + z_{it}$, $E[z] = 0$. 74 countries, 8670 observations, 1950-Q1–2026-Q2. Fitted countries: 74. Observations: 8670. Log-likelihood: 22608.76. Common intercept $a = -5.2386$ (0.0684). Common slope $g = 0.0079$ (0.0005).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.9769 (0.0864)	-0.7825 —	0.9710 (0.0581)
σ	0.0695 (0.0019)	0.0161 (0.0004)	0.0091 (0.0001)
ρ	0.9900 (0.0000)	0.9900 (0.0000)	0.9900 (0.0000)
$\Pi_0.$	0.7509 (0.0229)	0.1688 (0.0204)	0.0803 (0.0114)
$\Pi_1.$	0.0593 (0.0062)	0.9360 (0.0063)	0.0048 (0.0020)
$\Pi_2.$	0.0280 (0.0035)	0.0013 (0.0017)	0.9707 (0.0033)
π	0.1453 (0.0118)	0.3924 (0.0302)	0.4624 (0.0321)

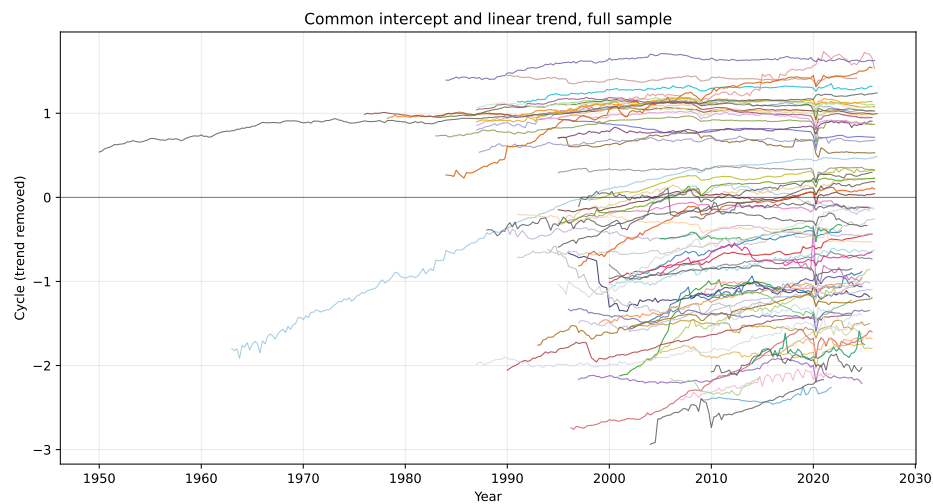


Figure 1: Country cycles after removing the common intercept and the linear trend.